

RADIOFREQUENCY AND PUBLIC HEALTH

The Impact of Radiofrequency Transmissions on Public Health

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Over the past several years, the potential health impacts from exposure to electromagnetic fields have gained widespread interest among the general public. Questions arose originally with regard to extremely low frequency electric and magnetic fields such as those found in the vicinity of electric-power lines. With the continued advancement of personal communications technology, similar concerns have arisen with regard to electromagnetic transmissions in the radiofrequency region, including that used in wireless communications. The following provides an overview of typical exposure levels and current information regarding federal exposure guidelines for radiofrequency electromagnetic waves in the frequency range utilized by PCS and cellular systems.

Biological effects associated with exposure to intense electromagnetic fields, generally the result of thermal heating, are dependent on two characteristics of electromagnetic waves, frequency and power. Using speech as a comparison, the frequency of an electromagnetic wave relates to the pitch at which one speaks, for example treble or bass, and power is described by how loud one speaks. The scientific unit used to describe frequency is the "Hertz", usually abbreviated Hz. The Hertz represents one repetition per second. For example, if one's heart beats 70 times per second, we could say that it beats at a frequency of 70 Hz. Or, a computer chip that operates at 1 GHz (gigahertz, giga meaning billion) does 1 billion operations per second. The figure below shows a variety of common uses within the electromagnetic spectrum. PCS, for example, operates in the 1.9 GHz frequency range.

The unit used to describe power is typically the Watt, W. The Watt describes the amount of energy transferred during one second. Most of us are familiar with this unit in our everyday lives because we use incandescent light-bulbs; a 100 W light bulb uses twice the energy of a 50 W light bulb. Again, as an example, PCS towers operate at a typical power output in each of three sectors of about 200 Watts.

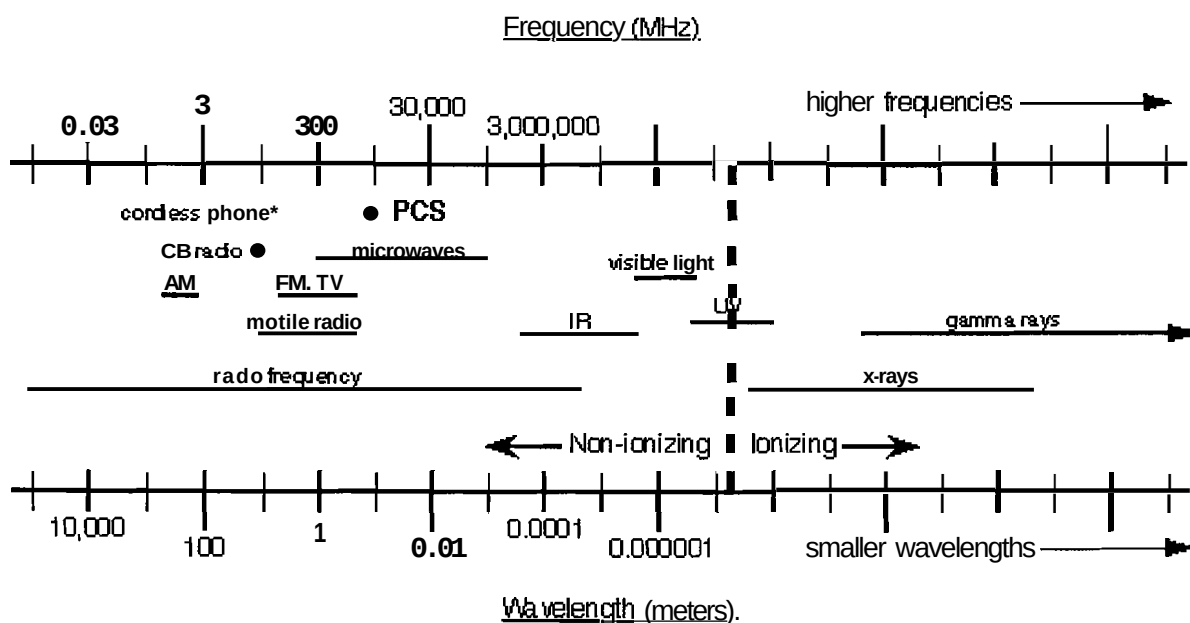
Radiofrequency exposure limits for members of the general public are set into law by the Federal Communications Commission (FCC) based on recommendation from the Institute of Electrical and Electronics Engineers (IEEE), the American National Standards Institute (ANSI), and the National Council on Radiation Protection and Measurements (NCRP). Today's standards identify the amount of energy per second (usually in milliWatts, where 1000 milliWatts equals 1 Watt) that can be transmitted through a square inch or, more typically, a square centimeter.

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Accordingly, exposure limits are expressed in terms of "power density" and are in units of mW/cm^2 (milliWatts per square centimeter). On August 1, 1996, the FCC suggested standardization of exposure limits and went with the NCRP recommendation of $1 \text{ mW}/\text{cm}^2$. Regulations within in the past couple of years in Australia and New Zealand stipulate identical limits to those of the NCRP, and regulations in the Netherlands allow exposure limits twice this value. The International Commission on Non-Ionizing Radiation Protection (ICNIRP) has also adopted the exposure limit of $1 \text{ mW}/\text{cm}^2$ for members of the general public at frequencies of 2 GHz and higher.

These limits can be put into perspective somewhat, by comparison with the metabolic heat production rate for an average human at rest, about $5 \text{ mW}/\text{cm}^2$. With increased activity, the metabolic heat production rate can naturally rise to as much as $80 \text{ mW}/\text{cm}^2$ depending on a number of physiological parameters such as age, gender, size, and general physical fitness.

ELECTROMAGNETIC SPECTRUM



As stated above, PCS and cellular systems emit electromagnetic waves at frequencies in the GHz range with a nominal power of about 200 Watts per sector. Under these conditions, the maximum power density at 100 feet laterally from the antenna element is calculated to be $0.010 \text{ mW}/\text{cm}^2$ or 100 times below the NCRP limit. This estimate assumes that the transmission power is emitted equally in all directions within the focused area (see power density calculation shown below). However, the bulk of the transmitted power is normally aimed only at the horizon, meaning that actual power levels on the ground, 100 feet from the base of the tower are much less than this calculated value. Power densities at this level are much lower than the federal limit and a great deal lower than levels where any type of biological effect has been observed. The

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heat production rate in our bodies caused by metabolism is thousands of times greater than any heat generated as a result of these transmissions.

Comparison of EMF Transmissions to Other RF Sources. To provide a better understanding of what these power densities mean, we can compare PCS transmissions with the power density levels of other well-known radiofrequency (RF) devices (see the table below). These include, but are not limited to, baby monitors, anti-theft monitors and automatic door openers at department stores, cordless telephones, **police/fire** two-way radios, and television and radio broadcasting stations. For example, standard television and radio towers are generally much taller than PCS or cellular towers, and also emit signals with power levels up to 5 million Watts. A 1,000 foot TV tower holding an antenna that emits 5 million Watts in all directions potentially has a power density at its base of 0.43 mW/cm^2 , more than 40% of the federal limit for emissions at that frequency. Baby monitor transmitters operate at a power of a few Watts. Therefore, the power density at a distance of 100 feet from a 200 Watt PCS transmitter would be equal to the power density at about four (4) feet from the baby monitor.

Some common radiofrequency sources.

Source	Frequency range (MHz)	Maximum power (Watts)
Anti-theft device	0.003 - 30	0.001
Automatic door opener	10,000	0.01
Hand-held cellular	800 - 900	0.6
Cordless phone	46 - 60	1
CB radio	27	5
Mobile radio	27 - 900	100
PCS	1,850 - 1,945	200
Microwave oven	915 - 2,450	2000
Satellite communications	100 - 275,000	5,000
AM Radio	0.535 - 1.705	50,000
FM Radio	88 - 108	200,000
LORAN transmitter	0.1	3,000,000
Television	54 - 890	5,000,000

To summarize, the calculated and measured power densities resulting from PCS transmissions are many times lower than the national (and international) RF standard of 1 mW/cm^2 . The most recent literature on the potential health effects of RF exposure continues to provide a vast majority of studies whose investigators find no statistical difference between exposed and non-exposed persons for various biological effects studied. A few investigators suggest that their data show associations between exposure and biological effect. These studies, however, are generally unable to provide a causal relationship between RF exposure and biological effect.

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Additionally, studies of effects on rats that are repeated are many times unable to reproduce results. One group of investigators attempted to reproduce their own results by conducting their experiment in triplicate. They showed one positive and two negative findings when looking at a particular biological effect in rats following exposure to radiofrequency radiation. This study, alone shows the variability in research of this type. Scientific studies have failed to demonstrate a human health effect as the result of exposure to low-level radiofrequency electromagnetic fields. The scientific data indicate that health risks from PCS or cellular RF transmissions are negligible.

Calculation of Maximum Power Density. Power density can be calculated at any distance from an electromagnetic source if the source is assumed to be emitted from a single point. The power density is simply the total power emitted divided by the area through which the power is directed. In the case of a typical cell tower, this area is, $4\pi r^2/6$, the surface-area of a spherical triangle with radius r , a horizontal angle of 120° and a vertical angle of approximately 90° . If the power output of the transmitter is 200 Watts (or 200,000 milliWatts), then given the shortest urban tower, i.e., a distance of 100 feet (3,048 centimeters), the power density is:

$$\frac{200,000 \cdot 6}{4 \cdot 3.14159 \cdot 3048^2} = 0.010 \text{ mW/cm}^2.$$

Because of the inverse-square principle, which applies to electromagnetic emission sources, every time the distance from the tower is doubled, the power density decreases four times. For example, we've shown that at 100 feet horizontally from the top of a cell tower the maximum power density is 0.010 mW/cm^2 (100 times lower than the RF standard). Based on the inverse-square principle we can estimate that at 200 feet away from the source the maximum power density would be four times lower or about 0.0025 mW/cm^2 (400 times lower than the RF standard) and at 400 feet the maximum power density is reduced to 0.00063 mW/cm^2 (1,600 times lower than the RF standard). Therefore, power density drops very rapidly as we move farther and farther away from the emission source.